In [1]:
```
import pandas as pd
import numpy as np

import warnings
warnings.filterwarnings('ignore')
```

In [2]:
```
df= pd.read_csv('diabetes.csv')
```

In [3]:
```
df
```

Out[3]:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	Diabete
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	
...	...	...	...	...	...	...	
763	10	101	76	48	180	32.9	
764	2	122	70	27	0	36.8	
765	5	121	72	23	112	26.2	
766	1	126	60	0	0	30.1	
767	1	93	70	31	0	30.4	

768 rows × 9 columns

In [4]:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

x= df.iloc[:,0:-1]
y=df.Outcome
```

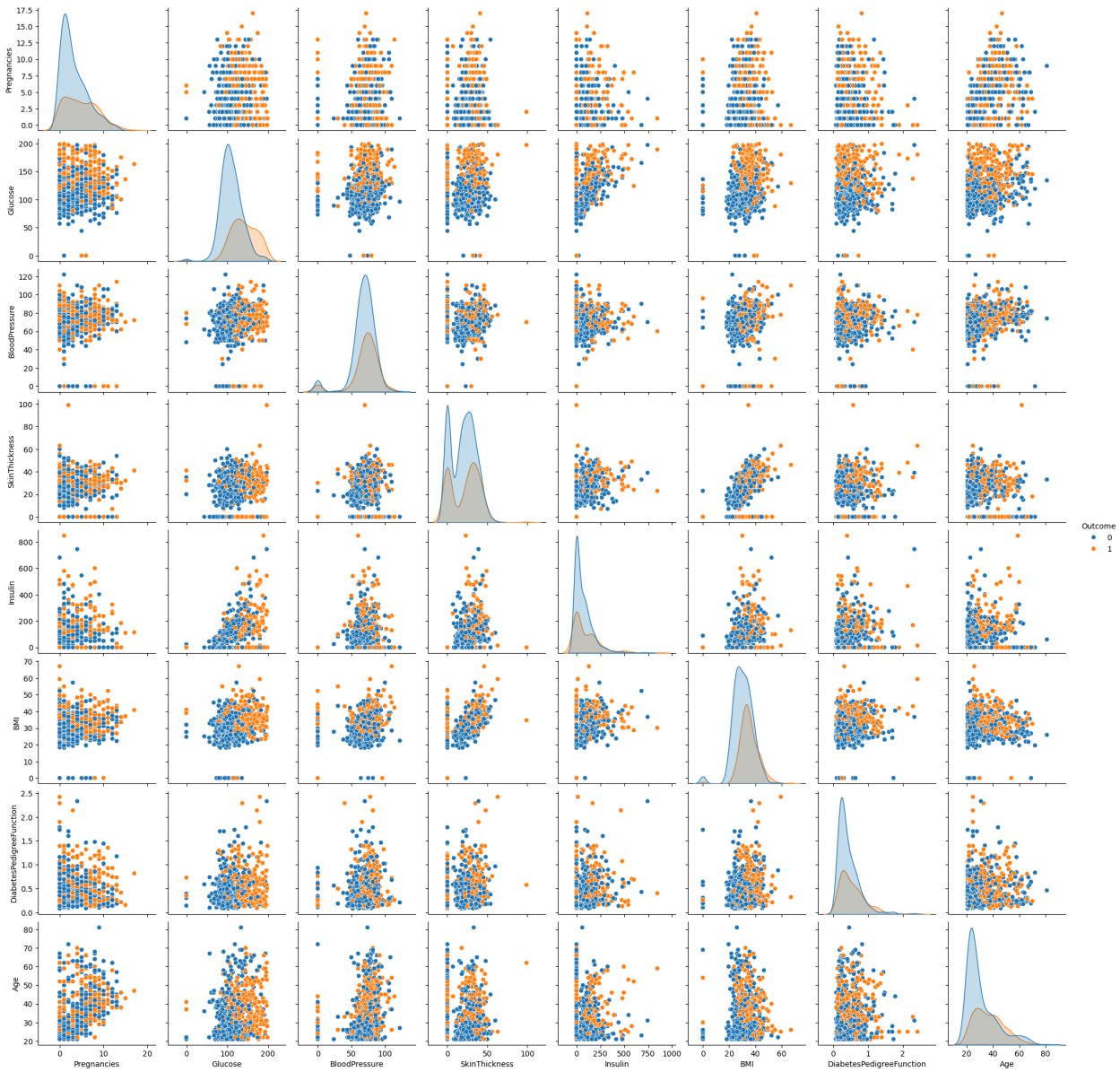
In [5]:
```
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.2,random_stat
```

In [6]:
```
print('rows for training',x_train.shape[0])
print('rows for testing',x_test.shape[0])
```

```
rows for training 614
rows for testing 154
```

In [7]:
```
import seaborn as sns
sns.pairplot(df,hue='Outcome')
```

Out[7]:
```
<seaborn.axisgrid.PairGrid at 0x1a5f29e9400>
```



In [8]:
```
lr =LogisticRegression()
lr
```

Out[8]:
```
▾ LogisticRegression
LogisticRegression()
```

In [9]:
```
# fit
lr.fit(x_train,y_train)
```

Out[9]:
```
▾ LogisticRegression
LogisticRegression()
```

In [10]:
```
#pred
```

```
y_pred = lr.predict(x_test)
```

In [11]:
```
y_pred
from sklearn.metrics import accuracy_score,confusion_matrix,precision_score
```

In [12]:
```
round(accuracy_score(y_test,y_pred),4)*100
```

Out[12]: 74.68

In [13]:
```
pd.DataFrame(confusion_matrix(y_test,y_pred))
```

Out[13]:

	0	1
0	78	21
1	18	37

In [14]:
```
tn=78
tp=37
fp=21
fn=18
```

In [15]:
```
#accuracy

(tp+tn)/(tp+tn+fp+fn)
```

Out[15]: 0.7467532467532467

In [16]:
```
(78+37)/(78+37+18+21)
```

Out[16]: 0.7467532467532467

In [17]:
```
precision_score(y_test,y_pred)
```

Out[17]: 0.6379310344827587

In [18]:
```
#precision
tp/(tp+fp)
```

Out[18]: 0.6379310344827587

In [19]:
```
37/(37+21)
```

Out[19]: 0.6379310344827587

In [ ]: